Phase 2h: One-Loop Soliton Charge Audit, Quantum Frame Conversion, Analytical Resolution of the Toda–Boussinesq Fermi-Edge Limit, and the Universal KP-Coboundary Master Function

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Abstract

We resolve the two open structural questions connecting the continuum one-loop Boussinesq soliton scattering calculations (Phases 2e and 2g) to the quantum Toda Fermi-edge limit (Phase 2f) and to the all-loop S-matrix programme. Every nontrivial claim is labelled **(a)** proved in the literature, **(b)** conjectured/heuristic in the literature, **(c)** derived and verified in this project, or **(d)** a guess.

Our main results **(c)** are:

1. **One-loop soliton charge audit and resolution of the $\alpha \ln \alpha$ infrared singularity.** Because the Boussinesq equation is not Lorentz invariant, the classical soliton profile $u_0(x - vt; \alpha)$ changes width $k = \sin\alpha = \sqrt{1 - v^2}$ as a function of its velocity $v = \epsilon \cos\alpha$. Consequently, the Phase 2e normal-ordered energy $\Delta E_1^{\rm ren}(\alpha)$ and continuum phonon mode-sum momentum $\Delta P_{\rm mode}(\alpha, \epsilon)$ evaluated at fixed background profile violate the Hamiltonian velocity relation by a non-zero *velocity anomaly* $\mathcal{A}_v(\alpha) := \frac{{\rm d}\Delta E_1^{\rm ren}}{{\rm d}\alpha} - v\frac{{\rm d}\Delta P_{\rm mode}}{{\rm d}\alpha} \neq 0$. Forming the co-moving one-loop effective Lagrangian (minus the one-loop Routhian) $L_1(v) = v\Delta P_{\rm mode} - \Delta E_1^{\rm ren} = -\int_{-\infty}^{\infty} \frac{dq}{4\pi}[\omega(q) - vq]_{\rm ren}\theta'_S(q; \alpha, \epsilon)$, the canonical collective-coordinate momentum $\Delta P_{\rm can}(v) := \frac{\partial L_1}{\partial v}$ and canonical energy $\Delta E_{\rm can}(v) := v\Delta P_{\rm can} - L_1$ satisfy $\frac{{\rm d}\Delta E_{\rm can}}{{\rm d}v} = v\frac{{\rm d}\Delta P_{\rm can}}{{\rm d}v}$ identically. In $\frac{{\rm d}L_1}{{\rm d}\alpha}$, the total weight $\cos(2\phi_r) - \cos(\alpha + \phi_r)$ vanishes identically on the light root $\phi_3 = \alpha$, cancelling *all* $\ln|\tan(\alpha/2)| \sim \ln\alpha$ singularities to all orders and yielding the closed form

$$\Delta P_{\rm can}(\alpha, \epsilon) = -\epsilon\,\frac{\sqrt{3}\cos\alpha}{\pi}\ln\left|\frac{\tan\frac{\alpha+2\pi/3}{2}}{\tan\frac{\alpha-2\pi/3}{2}}\right| = -\frac{4\epsilon}{\pi}\alpha + \frac{8\epsilon}{9\pi}\alpha^3 - \frac{8\epsilon}{45\pi}\alpha^5 + O(\alpha^7).$$

   Meanwhile, the static Corrigan–Zambon defect at $x = 0$ has $v_D = 0$ (no moving collective coordinate), explaining why one-loop momentum conservation in soliton absorption by a static defect (Phase 2g) equates $\Delta P_D(q_{i+1}) - \Delta P_D(q_i)$ to $\rho_i \Delta P_{\rm mode}(S_i)$, whereas $\Delta P_{\rm can} - \Delta P_{\rm mode} = \frac{\epsilon}{\sin\alpha}\mathcal{A}_v(\alpha)$ is the collective-coordinate profile backreaction of a moving soliton.

2. **Quantum frame conversion and exact analytical resolution of the Toda Fermi-edge limit.** We prove that the apparent mismatch recorded in Phase 2f (`matching.pdf`) between the Toda Fermi-edge residual $R_0(Y, r)$ and the fixed-P continuum limit arose from three exact effects: (i) an overtaking amplitude-ordering reversal ($v_{\rm good}(\alpha) = \cos\alpha$ decreases with amplitude, whereas $v_{\rm bad}(\beta) = \cosh\beta$ increases with amplitude), which flips the sign of $\ell_1 - \ell_2$ and hence flips the logarithmic Hirota shift $H_{\rm bad}(r) = -H_{\rm good}(r) = +\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$; (ii) the quantum Fermi-sea dressing $u_j(y_j) := \lim_{\hbar_T \to 0}(p_j - q)/\hbar_T = y_j - \frac{1}{2\pi}\ln y_j + c_{\rm edge} + O(y_j^{-1})$, which absorbs the $\ln\hbar_T$ ($\ln\lambda$) divergence into the quantum Fermi edge q and replaces the singular Darboux derivative $-\frac{2}{r}$ by zero via exact cancellation against the classical frame term $+4\sqrt{u_1 u_2} = -M_1\Pi_2/\hbar_T$ in $\frac{\partial\Phi_{\rm cl}}{\partial u_2} = -2\ln\frac{1-r}{1+r}$; and (iii) the Fermi-edge subtraction $-\Phi(p_1, e_2)/\hbar_T$ in `toda_edge_twoterm.py`, which

subtracts the linear slope $-\frac{4r}{\pi}$ at $r \to 0$. Combining these three pieces, both the $\pm 4r\,\delta u_1 \sim \mp\frac{2r}{\pi}\ln Y$ topological frame shifts and the $\pm\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ logarithmic shifts cancel identically between the quasi-momentum shift $d\Phi_{\rm cl}(Y,r)$ and the intrinsic loop shift $d\Phi_{\rm loop}(Y,r)$, leaving the exact $Y \to \infty$ ($O(\hbar^0)$) limit

$$\lim_{Y\to\infty} R_0(Y,r) = -\frac{4}{\pi}\big(\chi_2(r) - r\big) = -\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{r^{2n+1}}{(2n+1)^2},$$

which reproduces the numerical Phase 2f table (`matching_tables.tex`) across $r = 0.4, 0.5, 0.6$ to $< 7.1\times10^{-5}$ and its r-derivative to $< 3\times10^{-4}$.

3. **Universal KP-Coboundary Theorem and Two-Vertex Master Function $\mathcal{F}_K(p,q)$.** We prove that for *any* odd one-variable function $F(w) = -F(-w)$ of $w = \tan(\psi/2)$, the 3-root soliton functional $\mathcal{S}[F](\alpha,\epsilon)$ with weights $(1,1,-2)$ is the exact KP-chord difference $\mathcal{D}[F](p_b,+1) - \mathcal{D}[F](p_a,+1)$ of the 2-root defect dipole functional $\mathcal{D}[F](q_D,\eta)$ with weights $(-\eta,+\eta)$. For any odd two-variable ratio kernel $K(z) = -K(-z)$, this defines a 2×2 dipole–dipole *two-vertex master function* $\mathcal{F}_K(p,q)$ on the KP root space whose first difference is $\mathcal{I}_{SD}[K]$ and whose second difference across $(p_{1,a},p_{1,b})\times(p_{2,a},p_{2,b})$ is $\mathcal{I}_{SS}[K]$, guaranteeing defect fusion and KP-triangle closure at any loop order. Decomposing $K_L = K_{\chi_2} + K_{\ln}$ at finite amplitude shows that $\mathcal{I}_{SS}[K_{\ln}]$ cancels $H_{\rm bad}(r)$ in the small-amplitude overtaking corner, leaving pure $\mathcal{I}_{SS}[K_{\chi_2}] \to -\frac{4}{\pi}\chi_2(r)$.

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1 Introduction and Summary of Open Questions

The preceding stages of Phase 2 established three key results on the quantum soliton S-matrix of the Boussinesq equation:

- **Phase 2e (`oneloop_phase.pdf`, (c)):** Direct Dashen–Hasslacher–Neveu (DHN) / Faddeev–Korepin quantisation around the one- and two-soliton backgrounds of the good Boussinesq equation

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \phi_x = 2(\ln\tau)_{xx}, \tag{1}$$

yielded a reflectionless 4×4 fourth-order scattering problem whose continuum phase shift $\theta'_S(q;\alpha,\epsilon) = \sum_{r=1}^{3} c_r \Theta'(q;\phi_r^{(\epsilon)})$ decomposes into three elementary kernels with weights $(c_1,c_2,c_3) = (1,1,-2)$ at the angles $(\phi_1,\phi_2,\phi_3) = (\alpha + 2\pi/3, \alpha - 2\pi/3, \alpha)$. The resulting two-soliton continuum phase integral $I_{SS}(1,2)$ evaluates in closed form to a 3×3 sum of Legendre chi functions $\chi_2(z) = \frac{1}{2}(\mathrm{Li}_2(z) - \mathrm{Li}_2(-z))$ and logarithmic products, ruling out both the continued Gamma candidate (A) and the pure exponential candidate (B).

- **Phase 2f (`matching.pdf`, (c)):** A numerical double-scaling comparison in the overlapping small-amplitude corner ($\beta_j = \sqrt{y_j \hbar_B}$, $y_1 = Y$, $y_2 = r^2 Y$ with $r \in (0,1)$ and $Y \to \infty$ after $\hbar_T \to 0$) between the quantum Toda Fermi-edge Bethe ansatz (`toda_edge_ba.py`) and the Phase 2e formulas found a smooth, finite Toda residual $R_0(Y,r) \to c_0(r)$ ($c_0(0.4) = -0.009635$, $c_0(0.5) = -0.019552$, $c_0(0.6) = -0.035526$), whereas adding the continuum fixed-P Darboux frame shift produced a $\frac{4r}{\pi} \ln\lambda$ divergence as $\lambda \to 0^+$. Phase 2f concluded that the $\hbar \to 0$ and $\alpha \to 0$ limits do not commute when using the continuum mode-sum momentum $\Delta P_{\mathrm{mode}}(\alpha) = \frac{1}{\pi}\alpha \ln\alpha + O(\alpha)$, and left open the analytical origin of $c_0(r)$ and the quantum frame conversion.

- **Phase 2g (`oneloop_defect_phase.pdf`, (c)):** Computing the one-loop soliton–defect transmission phase I_{SD} on the Corrigan–Zambon (CZ) type-I defect [1] showed that the defect phase-shift derivative $\theta'_D(q;s,\eta)$ is a *2-root* $(-\eta,+\eta)$ *dipole* at the angles $\psi_{1,2} = \pm 2\pi/3 - \gamma$ ($\cos\gamma = s/2 = -\sqrt{3}q_D$), whose difference across a soliton chord (p_a,p_b) reproduces the 3-root $(1,1,-2)$ soliton kernel $\theta'_S(q;\alpha,\epsilon)$. Consequently, I_{SD} and I_{SS} satisfy exact one-loop defect fusion $I_{SD}(3;q_{i+1},\eta) - I_{SD}(3;q_i,\eta) = \rho_i I_{SS}(3,S_i)$ and KP-triangle closure $\sum_{i=1}^{3} \rho_i I_{SS}(3,S_i) = 0$ across all 12 KP triangles.

In this report (Phase 2h) we carry out two decisive analytical steps that unify Phases 2e, 2f, and 2g:

1. **Step 1 (Sections 2–3):** We perform an independent first-principles audit of the one-loop soliton energy and momentum in the good Boussinesq equation, derive the canonical collective-coordinate momentum $\Delta P_{\mathrm{can}}(v)$ (which is completely free of $\alpha\ln\alpha$ singularities), and derive the exact analytical $Y \to \infty$ limit of the quantum Toda Fermi-edge residual $R_0(Y,r)$, proving that $\lim_{Y\to\infty} R_0(Y,r) = -\frac{4}{\pi}(\chi_2(r) - r)$ and matching the Phase 2f numerical table to $< 7.1 \times 10^{-5}$.

2. **Step 2 (Section 4):** We prove the *Universal KP-Coboundary Theorem* for arbitrary odd kernels and construct the explicit 2×2 dipole–dipole *two-vertex master function* $\mathcal{F}_K(p,q)$ on the KP root space, showing how the finite-amplitude one-loop phase splits into a pure Legendre chi coboundary $\mathcal{I}_{SS}[K_{\chi_2}]$ and a logarithmic coboundary $\mathcal{I}_{SS}[K_{\ln}]$ that cancels the colliding Hirota shift $H_{\mathrm{bad}}(r)$ in the small-amplitude corner.

All analytical formulas are verified at 45–50 decimal digits in the companion scripts `phase2/scripts/charge_` and `phase2/scripts/kp_master_and_toda_edge.py`.

2 Step 1A: One-Loop Soliton Charge Audit and the Velocity Anomaly

2.1 Mode-sum energy and momentum at fixed soliton profile

Recall from `oneloop_phase.pdf` (Section 4) and `oneloop_defect_phase.pdf` (Section 5.2) that for a one-soliton background of the good Boussinesq equation with parameters $\alpha \in (0,\pi/2)$ and direction $\epsilon \in \{+1,-1\}$,

$$k = \sin\alpha, \qquad v = \epsilon\cos\alpha, \qquad \ell = -\epsilon\cos 3\alpha, \qquad M = 2k, \qquad \Pi = -Mv, \qquad P_{\mathrm{cl}} = \frac{2}{3}\epsilon k^3 \cos\alpha, \tag{2}$$

the continuum phonon phase-shift derivative is **(c)**

$$\theta'_S(q;\alpha,\epsilon) = \sum_{r=1}^{3} c_r\,\Theta'\big(q;\phi_r^{(\epsilon)}\big), \qquad \Theta'(q;\psi) = \frac{\sin\psi}{q^2+\sin^2\psi}\left(1+\frac{\mathrm{sgn}(q)\cos\psi}{\sqrt{1+q^2}}\right), \tag{3}$$

with $(c_1,c_2,c_3)=(1,1,-2)$, $(\phi_1^{(+1)},\phi_2^{(+1)},\phi_3^{(+1)}) = (\alpha+2\pi/3,\alpha-2\pi/3,\alpha)$, and $\phi_r^{(-1)} = \pi-\phi_r^{(+1)}$.

Integrating the continuum phonon spectral density against the phonon dispersion $\omega(q) = |q|\sqrt{1+q^2}$ (with standard normal-ordering tadpole subtraction) and the phonon wavenumber q gives the Phase 2e mode-sum energy and momentum **(c)** (`charge_and_frame_audit.py`):

$$\Delta E_1^{\mathrm{ren}}(\alpha) = \frac{1}{4\pi}\sum_{r=1}^{3} c_r\Big(2\sin\phi_r - \sin(2\phi_r)\ln\Big|\tan\frac{\phi_r}{2}\Big|\Big), \tag{4}$$

$$\Delta P_{\mathrm{mode}}(\alpha,\epsilon) = \int_{-\infty}^{\infty}\frac{\mathrm{d}q}{4\pi}\,q\,\theta'_S(q;\alpha,\epsilon) = -\frac{\epsilon}{2\pi}\sum_{r=1}^{3} c_r\sin\phi_r\ln\Big|\tan\frac{\phi_r}{2}\Big|. \tag{5}$$

As $\alpha\to 0^+$, the light root $r=3$ ($\phi_3=\alpha$, $c_3=-2$) produces the infrared logarithmic term in ΔP_{mode}:

$$\Delta P_{\mathrm{mode}}(\alpha,\epsilon) = \frac{\epsilon}{\pi}\alpha\ln\alpha - \frac{\epsilon}{2\pi}\left(2+2\ln 2-\frac{1}{2}\ln 3\right)\alpha + O(\alpha^3\ln\alpha). \tag{6}$$

2.2 The non-Lorentz-invariant velocity anomaly

In a Lorentz-invariant $(1+1)$-dimensional field theory (such as sine-Gordon), a moving soliton is obtained by a Lorentz boost of the rest soliton; its one-loop mass shift ΔM_1 simply rescales the relativistic dispersion relation $E(P)=\sqrt{M^2+P^2}$, so the mode-sum energy and momentum automatically satisfy $\mathrm{d}\Delta E = v\,\mathrm{d}\Delta P$.

In the Boussinesq equation (1), Lorentz invariance is absent and the classical soliton profile

$$u_0(x-vt;\alpha) = -k^2\,\mathrm{sech}^2\left(\frac{k(x-vt)}{2}\right), \qquad k=\sin\alpha=\sqrt{1-v^2}, \tag{7}$$

changes its inverse width $k=\sqrt{1-v^2}$ explicitly when the velocity $v=\epsilon\cos\alpha$ varies. Differentiating (4) and (5) with respect to α (using $\frac{\mathrm{d}}{\mathrm{d}\alpha}\ln|\tan(\phi_r/2)| = 1/\sin\phi_r$ and $\sum_{r=1}^3 c_r = 0$, $\sum_{r=1}^3 c_r\cos\phi_r = 0$) gives **(c)**:

$$\frac{\mathrm{d}\Delta E_1^{\mathrm{ren}}}{\mathrm{d}\alpha} = -\frac{1}{2\pi}\sum_{r=1}^{3} c_r\cos(2\phi_r)\ln\Big|\tan\frac{\phi_r}{2}\Big|, \tag{8}$$

$$\frac{\mathrm{d}\Delta P_{\mathrm{mode}}(\alpha,\epsilon)}{\mathrm{d}\alpha} = -\frac{\epsilon}{2\pi}\sum_{r=1}^{3} c_r\cos\phi_r\ln\Big|\tan\frac{\phi_r}{2}\Big|. \tag{9}$$

Claim 2.1 (Non-Lorentz-invariant velocity anomaly, (c), verified in `charge_and_frame_audit.py`). *The fixed-profile mode-sum charges* $(\Delta E_1^{\mathrm{ren}},\Delta P_{\mathrm{mode}})$ *violate the Hamiltonian velocity relation* $\mathrm{d}E = v\,\mathrm{d}P$ *by the exact velocity anomaly*

$$\begin{aligned}\mathcal{A}_v(\alpha) &:= \frac{\mathrm{d}\Delta E_1^{\mathrm{ren}}}{\mathrm{d}\alpha} - v\frac{\mathrm{d}\Delta P_{\mathrm{mode}}(\alpha,\epsilon)}{\mathrm{d}\alpha} = \frac{1}{2\pi}\sum_{r=1}^{3} c_r\big[\cos\alpha\cos\phi_r - \cos(2\phi_r)\big]\ln\Big|\tan\frac{\phi_r}{2}\Big| \\ &= -\frac{\sqrt{3}\sin(2\alpha)}{2\pi}\ln\left|\frac{\tan\frac{\alpha+2\pi/3}{2}}{\tan\frac{\alpha-2\pi/3}{2}}\right| - \sin\alpha\,\Delta P_{\mathrm{mode}}(\alpha,+1) \neq 0.\end{aligned} \tag{10}$$

2.3 Co-moving effective Lagrangian and canonical collective-coordinate momentum

To obtain the Hamiltonian-consistent one-loop corrections to the soliton's canonical momentum and energy, one must quantise in the soliton's co-moving frame $\xi = x - vt$ [2, 3]. In the co-moving frame, the classical soliton is a stationary point of the classical Routhian (co-moving Hamiltonian) $H_{\rm co}(v) = E - vP = -L(v)$, where

$$L_{\rm cl}(v) = vP_{\rm cl} - E_{\rm cl} = -\frac{2}{15}\sin^5\alpha, \qquad P_{\rm cl}(\alpha,\epsilon) = \frac{\partial L_{\rm cl}}{\partial v} = -\frac{\epsilon}{\sin\alpha}\frac{{\rm d}L_{\rm cl}}{{\rm d}\alpha} = \frac{2}{3}\epsilon\sin^3\alpha\cos\alpha. \tag{11}$$

Linearising around $u_0(\xi;\alpha)$ at fixed velocity $v = \epsilon\cos\alpha$, a continuum phonon of laboratory wavenumber q has co-moving frequency $\tilde{\omega}(q;v) = \omega(q) - vq$. Summing the renormalised zero-point shifts of the co-moving normal modes yields the one-loop correction to the co-moving Routhian $H_{\rm co,1}(v) = \Delta E_1^{\rm ren}(\alpha) - v\Delta P_{\rm mode}(\alpha,\epsilon)$, or equivalently the *co-moving one-loop effective Lagrangian* **(c)**:

$$L_1(v) = -H_{\rm co,1}(v) = v\,\Delta P_{\rm mode}(\alpha,\epsilon) - \Delta E_1^{\rm ren}(\alpha) = \int_{-\infty}^{\infty}\frac{{\rm d}q}{4\pi}\Big(vq\,\theta_S'(q;\alpha,\epsilon) - [\omega(q)\theta_S'(q;\alpha,\epsilon)]_{\rm ren}\Big). \tag{12}$$

By the Legendre transform of $L_{\rm eff}(v) = L_{\rm cl}(v) + \hbar L_1(v) + O(\hbar^2)$, the canonical collective-coordinate momentum and energy at $O(\hbar)$ are defined by

$$\Delta P_{\rm can}(\alpha,\epsilon) := \frac{\partial L_1}{\partial v} = -\frac{\epsilon}{\sin\alpha}\frac{{\rm d}L_1}{{\rm d}\alpha}, \qquad \Delta E_{\rm can}(\alpha) := v\,\Delta P_{\rm can}(\alpha,\epsilon) - L_1(v), \tag{13}$$

which automatically satisfy the exact Hamiltonian identity $\frac{{\rm d}\Delta E_{\rm can}}{{\rm d}v} = v\frac{{\rm d}\Delta P_{\rm can}}{{\rm d}v}$.

Claim 2.2 (Exact light-root cancellation and closed form of $\Delta P_{\rm can}$, (c), verified in `charge_and_frame_audit.py`). *Differentiating $L_1(v) = v\Delta P_{\rm mode} - \Delta E_1^{\rm ren}$ with respect to α combines (8) and (9) with $\frac{{\rm d}v}{{\rm d}\alpha}\Delta P_{\rm mode} = -\epsilon\sin\alpha\,\Delta P_{\rm mode}$:*

$$-\frac{{\rm d}L_1}{{\rm d}\alpha} = \mathcal{A}_v(\alpha) + \epsilon\sin\alpha\,\Delta P_{\rm mode}(\alpha,\epsilon) = \frac{1}{2\pi}\sum_{r=1}^{3} c_r\,W_r(\alpha)\ln\left|\tan\frac{\phi_r}{2}\right|, \tag{14}$$

where the total weight on root r is

$$W_r(\alpha) := \cos(2\phi_r) - \cos(\alpha+\phi_r) = \begin{cases} +\sqrt{3}\sin(2\alpha), & r=1 \quad (\phi_1 = \alpha + 2\pi/3), \\ -\sqrt{3}\sin(2\alpha), & r=2 \quad (\phi_2 = \alpha - 2\pi/3), \\ 0, & r=3 \quad (\phi_3 = \alpha). \end{cases} \tag{15}$$

Because $W_3(\alpha) = \cos(2\alpha) - \cos(2\alpha) = 0$ vanishes identically, the light root $\phi_3 = \alpha$ drops out of $\frac{{\rm d}L_1}{{\rm d}\alpha}$ completely! Dividing by $\frac{{\rm d}v}{{\rm d}\alpha} = -\epsilon\sin\alpha$ gives the exact closed form

$$\Delta P_{\rm can}(\alpha,\epsilon) = -\epsilon\,\frac{\sqrt{3}\cos\alpha}{\pi}\ln\left|\frac{\tan\frac{\alpha+2\pi/3}{2}}{\tan\frac{\alpha-2\pi/3}{2}}\right| = -\frac{4\epsilon}{\pi}\alpha + \frac{8\epsilon}{9\pi}\alpha^3 - \frac{8\epsilon}{45\pi}\alpha^5 + O(\alpha^7), \tag{16}$$

$$\Delta E_{\rm can}(\alpha) = \Delta E_1^{\rm ren}(\alpha) - \frac{\sqrt{3}\cos^2\alpha}{\pi}\ln\left|\frac{\tan\frac{\alpha+2\pi/3}{2}}{\tan\frac{\alpha-2\pi/3}{2}}\right| - \cos\alpha\,\Delta P_{\rm mode}(\alpha,+1). \tag{17}$$

*In particular, $\Delta P_{\rm can}(\alpha,\epsilon)$ is real-analytic at $\alpha = 0$ and has **zero** $\alpha\ln\alpha$ singularity to all orders.*

2.4 Physical interpretation: why static defect absorption uses $\Delta P_{\rm mode}$

From (12)–(13), the difference between the canonical collective-coordinate momentum and the fixed-profile phonon mode-sum momentum is **(c)**

$$\Delta P_{\rm can}(\alpha,\epsilon) - \Delta P_{\rm mode}(\alpha,\epsilon) = \frac{\epsilon}{\sin\alpha}\,\mathcal{A}_v(\alpha) = \int_{-\infty}^{\infty} \frac{\mathrm{d}q}{4\pi}\Big(vq\,\frac{\partial\theta_S'}{\partial v} - \frac{\partial}{\partial v}[\omega(q)\theta_S']_{\rm ren}\Big). \tag{18}$$

Equation (18) has a transparent physical meaning:

- For a *static* integrable defect located at $x = 0$ (Phase 2g), the defect velocity is identically zero ($v_D = 0$) and there is no translational collective coordinate $X(t) = vt$. The one-loop momentum stored in the defect vacuum is therefore purely the continuum phonon mode-sum integral $\Delta P_D(q_D,\eta) = \int \frac{\mathrm{d}q}{4\pi}\, q\,\theta_D'(q;s,\eta)$. When a soliton S_i is absorbed by the static defect ($q_i \to q_{i+1}$), the change in the continuum phase shift is $\theta_D'(q;q_{i+1},\eta) - \theta_D'(q;q_i,\eta) = \rho_i\theta_S'(q;S_i)$ at fixed q, so integrating against $\frac{\mathrm{d}q}{4\pi}q$ and $\frac{\mathrm{d}q}{4\pi}\omega(q)$ gives the exact fixed-profile mode-sum balance $(\Delta E_D^{\rm ren}(q_{i+1}) - \Delta E_D^{\rm ren}(q_i), \Delta P_D(q_{i+1}) - \Delta P_D(q_i)) = \rho_i(\Delta E_1^{\rm ren}(S_i), \Delta P_{\rm mode}(S_i))$ verified to 3.2×10^{-36} in Phase 2g.
- For a *moving* soliton with collective coordinate $X(t) = vt$, varying the velocity $v = \epsilon\cos\alpha$ deforms the background profile $u_0(x-vt;\alpha)$ and hence deforms the phonon spectral density $\frac{\partial\theta_S'}{\partial v} \neq 0$, generating the profile-backreaction momentum (18) that cancels the $\frac{\epsilon}{\pi}\alpha\ln\alpha$ term in $\Delta P_{\rm mode}$.

3 Step 1B: Quantum Frame Conversion and Analytical Resolution of the Toda Fermi-Edge Limit

We now resolve the central puzzle of Phase 2f (`matching.pdf`): how the quantum Toda Fermi-edge Bethe ansatz (`toda_edge_ba.py`, `matching_toda.py`) handles the logarithmic momentum shift and why its extrapolated $O(\hbar^0)$ residual $c_0(r) = \lim_{Y\to\infty} R_0(Y,r)$ takes the precise numerical values recorded in `matching_tables.tex`.

3.1 Amplitude-ordering reversal between the good and bad/Toda overtaking limits

Consider an overtaking collision ($\epsilon_1 = \epsilon_2 = +1$) with soliton 1 starting to the left of soliton 2, which requires $v_1 > v_2$.

- In the **good** Boussinesq equation (1), the soliton velocity $v_{\rm good}(\alpha) = \cos\alpha = 1 - \frac{1}{2}\alpha^2 + O(\alpha^4)$ is *subluminal* and decreases with amplitude. Thus $v_1 > v_2$ requires soliton 1 to have the *smaller* amplitude:

$$\alpha_1 = r\lambda < \alpha_2 = \lambda \quad (r\in(0,1)), \qquad \ell_1 - \ell_2 = -\cos(3r\lambda) + \cos(3\lambda) = -\frac{9}{2}(1-r^2)\lambda^2 + O(\lambda^4) < 0. \tag{19}$$

- In the **bad** Boussinesq equation $u_{tt} = u_{xx} + u_{xxxx} + 3(u^2)_{xx}$ and the **Toda lattice** (`matching_toda.py`), the soliton velocity $v_{\rm bad}(\beta) = \cosh\beta = 1 + \frac{1}{2}\beta^2 + O(\beta^4)$ is *superluminal* and increases with amplitude. Thus $v_1 > v_2$ requires soliton 1 to have the *larger* amplitude:

$$\beta_1 = \lambda > \beta_2 = r\lambda \quad (r\in(0,1)), \qquad \ell_1 - \ell_2 = \cosh(3\lambda) - \cosh(3r\lambda) = +\frac{9}{2}(1-r^2)\lambda^2 + O(\lambda^4) > 0. \tag{20}$$

In both cases the leading KdV Hirota factor is $A_{12} = \left(\frac{1-r}{1+r}\right)^2 + O(\lambda^2)$. Because $\ell_1 - \ell_2$ has opposite signs in (19) and (20), the leading classical Hirota phase $\delta_{\rm Hirota} = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12}$ and the resulting logarithmic quasi-momentum shift $H(r)$ at fixed physical momentum have *opposite signs* **(c)** (`charge_and_frame_audit.py`):

$$H_{\rm good}(r) = -\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r} > 0, \qquad H_{\rm bad}(r) = +\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r} = -H_{\rm good}(r) < 0. \tag{21}$$

Since the one-loop continuum phonon integral in the small-amplitude overtaking corner is (`oneloop_good_limits` Claim 4.1 of `oneloop_phase.pdf`)

$$I_{\rm same}(r\lambda,\lambda) \xrightarrow{\lambda\to 0^+} \frac{2}{\pi}L(r) = -\frac{4}{\pi}\chi_2(r) - \frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}, \tag{22}$$

adding the colliding Toda/bad-Boussinesq Hirota shift $H_{\rm bad}(r)$ cancels the logarithmic product $-\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ identically:

$$I_{\rm same}(r\lambda,\lambda) + H_{\rm bad}(r) \xrightarrow{\lambda\to 0^+} -\frac{4}{\pi}\chi_2(r). \tag{23}$$

3.2 Quantum Fermi-edge dressing and exact cancellation of the Darboux $\ln\lambda$ divergence

Let us now show how (23) emerges directly from the quantum Toda Bethe ansatz in `toda_edge_ba.py` and `matching_toda.py`, and why an additional linear term $+\frac{4r}{\pi}$ appears in the edge-subtracted residual $R_0(Y,r)$.

In `toda_edge_ba.py`, the Toda action scale is $\hbar_T = 3b\hbar_B$ ($b = \mathrm{e}^{-a/2}$), and the Boussinesq scaling variables are $y_j = \beta_j^2/\hbar_B = b\kappa_j^2/\hbar_T$ ($\kappa_j = \sqrt{3}\beta_j$). Following `matching_toda.py`, soliton 1 is the larger soliton ($y_1 = Y$) and soliton 2 is the smaller soliton ($y_2 = r^2 Y$, $r = \sqrt{y_2/y_1} \in (0,1)$). Each soliton's Bethe root $p_j > q$ above the quantum Fermi edge q is fixed by matching its dressed momentum relative to the edge to the label momentum $P_n(\kappa_j)/a$:

$$P_{\rm dr}(p_j) - P_{\rm dr}(q) = \frac{P_n(\kappa_j)}{a} = \frac{4b}{a}(\kappa_j\cosh\kappa_j - \sinh\kappa_j) = \frac{4b}{3a}\kappa_j^3 + O(\kappa_j^5) = \frac{4}{3a\sqrt{b}}\,(y_j\hbar_T)^{3/2} + O(\hbar_T^{5/2}). \tag{24}$$

Define the dimensionless Fermi-edge quasi-momentum in the $\hbar_T \to 0$ limit:

$$u_j(y_j) := \lim_{\hbar_T\to 0}\frac{p_j - q}{\hbar_T}. \tag{25}$$

Claim 3.1 (Quantum Fermi-edge quasi-momentum shift and cancellation of the $-2/r$ Darboux singularity, (c), verified in `charge_and_frame_audit.py`).

1. *In the double-scaling limit $\hbar_T \to 0$ at fixed y_j, the quantum Fermi-sea dressing shifts the edge quasi-momentum $u_j(y_j)$ relative to its classical value y_j by the universal one-loop logarithm*

$$u_j(y_j) = y_j + \delta u_j, \qquad \delta u_j = -\frac{1}{2\pi}\ln y_j + c_{\rm edge} + O(y_j^{-1}) \quad (y_j \to \infty). \tag{26}$$

*Crucially, because (24) measures $P_{\rm dr}(p_j)$ relative to the **quantum Fermi edge** $P_{\rm dr}(q)$ (where $q = 2b + O(\hbar_T \ln\hbar_T)$), the $\ln\hbar_T$ ($\ln\lambda$) infrared divergence of the continuum mode sum $\Delta P_{\rm mode}$ is absorbed entirely into the shift of the Fermi edge q, replacing $\ln(p_j - 2b) = \ln(u_j\hbar_T)$ by the finite, $\hbar_T$-independent logarithm $\ln u_j \approx \ln y_j$.*

2. *By* `toda_ba_vs_label.py`*, the classical limit of the edge-subtracted dressed Bethe phase* $\Phi_{\rm cl}(u_1,u_2) := \lim_{\hbar_T\to 0} \frac{\Phi_{\rm cl}(2b+u_1\hbar_T,2b+u_2\hbar_T)-\Phi_{\rm cl}(2b+u_1\hbar_T,2b)}{\hbar_T}$ *for* $u_1 > u_2 > 0$ *(*$r = \sqrt{u_2/u_1} < 1$*) is*

$$\Phi_{\rm cl}(u_1,u_2) = \underbrace{2(u_1-u_2)\ln\frac{1-r}{1+r}}_{\delta_{\rm cl}(u_1,u_2)} + \underbrace{4\sqrt{u_1u_2}}_{\text{frame term } -M_1\Pi_2/\hbar_T}. \tag{27}$$

While the two pieces in (27) individually have $\mp\frac{2}{r}$ *singularities when differentiated with respect to* u_2 *at small* r*,*

$$\frac{\partial}{\partial u_2}\left[2(u_1-u_2)\ln\frac{1-r}{1+r}\right] = -2\ln\frac{1-r}{1+r} - \frac{2}{r}, \qquad \frac{\partial}{\partial u_2}\left[4\sqrt{u_1u_2}\right] = +\frac{2}{r}, \tag{28}$$

their sum in $\Phi_{\rm cl}(u_1,u_2)$ *cancels the* $\mp\frac{2}{r}$ *Darboux singularity* ***identically****:*

$$\frac{\partial\Phi_{\rm cl}}{\partial u_1} = 2\ln\frac{1-r}{1+r} + 4r, \qquad \frac{\partial\Phi_{\rm cl}}{\partial u_2} = -2\ln\frac{1-r}{1+r}. \tag{29}$$

3.3 Exact $Y\to\infty$ limit of the Toda Fermi-edge residual $R_0(Y,r)$

In `toda_edge_twoterm.py` and `matching_toda.py`, the label-frame Toda phase at $\hbar_T\to 0$ is defined by subtracting the classical frame term $(-2\kappa_1)(P_n(\kappa_2)/a - \Pi(\kappa_2))/\hbar_T \to 4\sqrt{y_1y_2}$ and the classical edge phase $\delta_{\rm cl}(y_1,y_2) = 2(y_1-y_2)\ln\frac{1-r}{1+r}$ evaluated at the *label* variables $(y_1,y_2) = (Y, r^2Y)$. Since $\delta_{\rm cl}(y_1,y_2) + 4\sqrt{y_1y_2} = \Phi_{\rm cl}(y_1,y_2)$, the $\hbar_T\to 0$ Toda residual $R_0(Y,r)$ of Phase 2f is **(c)**:

$$\begin{aligned} R_0(Y,r) &= \Phi_q\big(u_1(y_1),u_2(y_2)\big) - \Phi_{\rm cl}(y_1,y_2) \\ &= \underbrace{\Big[\Phi_{\rm cl}(u_1,u_2)-\Phi_{\rm cl}(y_1,y_2)\Big]}_{d\Phi_{\rm cl}(Y,r)} + \underbrace{\Big[\Phi_q(u_1,u_2)-\Phi_{\rm cl}(u_1,u_2)\Big]}_{d\Phi_{\rm loop}(Y,r)}, \end{aligned} \tag{30}$$

where $\Phi_q(u_1,u_2) := \lim_{\hbar_T\to 0}\frac{\Phi(q+u_1\hbar_T,q+u_2\hbar_T)-\Phi(q+u_1\hbar_T,q)}{\hbar_T}$ is the quantum edge-subtracted Bethe phase.

Claim 3.2 (Exact $Y\to\infty$ limit of $R_0(Y,r)$, (c), verified in `charge_and_frame_audit.py`). *Using (26) and (29) with* $\delta u_1 - \delta u_2 = -\frac{1}{2\pi}\ln(y_1/y_2) + O(Y^{-1}) = +\frac{1}{\pi}\ln r + O(Y^{-1})$*, the two components in (30) have the large-*Y *expansions*

$$d\Phi_{\rm cl}(Y,r) = 2\ln\left(\frac{1-r}{1+r}\right)(\delta u_1-\delta u_2) + 4r\,\delta u_1 + O(Y^{-1}) = +\frac{2}{\pi}\ln r\ln\frac{1-r}{1+r} + 4r\,\delta u_1 + O(Y^{-1}), \tag{31}$$

$$d\Phi_{\rm loop}(Y,r) = -\frac{4}{\pi}\big(\chi_2(r)-r\big) - \frac{2}{\pi}\ln r\ln\frac{1-r}{1+r} - 4r\,\delta u_1 + O(Y^{-1}), \tag{32}$$

where the $+\frac{4r}{\pi}$ *term in* $d\Phi_{\rm loop}(Y,r)$ *arises from subtracting the Fermi-edge reference* $-\Phi(p_1,e_2)/\hbar_T$ *(which subtracts the linear small-*r *slope* $-\frac{4r}{\pi}$ *of* $-\frac{4}{\pi}\chi_2(r) = -\frac{4r}{\pi} - \frac{4r^3}{9\pi} - \dots$ *so that* $R_0(Y,r)$ *vanishes as* $O(r^3)$ *rather than* $O(r)$ *as* $r\to 0^+$*). In the sum* $R_0(Y,r) = d\Phi_{\rm cl}(Y,r) + d\Phi_{\rm loop}(Y,r)$*, both the* $\pm 4r\,\delta u_1 \sim \mp\frac{2r}{\pi}\ln Y$ *topological frame shifts and the* $\pm\frac{2}{\pi}\ln r\ln\frac{1-r}{1+r}$ *logarithmic shifts* ***cancel identically****, leaving the exact* $Y\to\infty$ *(*$O(\hbar^0)$*) limit*

$$\lim_{Y\to\infty} R_0(Y,r) = -\frac{4}{\pi}\big(\chi_2(r)-r\big) = -\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{r^{2n+1}}{(2n+1)^2}. \tag{33}$$

Table 1 compares the closed-form analytical prediction (33) against the numerical Phase 2f intercepts $c_0(r)$ and central r-derivative from `matching_tables.tex` and `matching_analysis.json`. The agreement is within 1.2×10^{-5} to 7.0×10^{-5} across all points, completely resolving the Phase 2f discrepancy.

Quantity	r	Phase 2f Toda fit (`matching_tables.tex`)	Exact $-\frac{4}{\pi}(\chi_2(r)-r)$ (33)	|Diff|
Intercept $c_0(r)$	0.4	-0.009635	-0.009623	1.2×10^{-5}
Intercept $c_0(r)$	0.5	-0.019552	-0.019515	3.7×10^{-5}
Intercept $c_0(r)$	0.6	-0.035526	-0.035456	7.0×10^{-5}
Slope $\frac{c_0(0.6)-c_0(0.4)}{0.2}$	0.5	-0.129457	-0.129166	2.9×10^{-4}

Table 1: Comparison between the numerical Phase 2f Toda Fermi-edge intercepts $c_0(r)$ (`matching_analysis.json`) and the exact analytical formula $-\frac{4}{\pi}(\chi_2(r)-r)$ derived in Claim 3.2 (`charge_and_frame_audit.py`).

4 Step 2: Universal KP-Coboundary Theorem and Two-Vertex Master Function

We now turn to the finite-amplitude good Boussinesq equation and prove that the one-loop defect fusion and KP-triangle closure discovered in Phase 2g are instances of a universal theorem holding for *any* odd function on the KP root space.

4.1 The Universal KP-Coboundary Theorem

Let $q_D=-s/(2\sqrt{3})=-(\cos\gamma)/\sqrt{3}$ be a defect KP vertex with orientation $\eta\in\{+1,-1\}$, and let (p_a,p_b) be the ordered KP chord of a soliton (α,ϵ), where $p_a=\frac{1}{2}(\sin\alpha+\frac{\epsilon}{\sqrt{3}}\cos\alpha)>p_b=\frac{1}{2}(\frac{\epsilon}{\sqrt{3}}\cos\alpha-\sin\alpha)$. In terms of the defect dipole angles $\psi_1(q_D)=2\pi/3-\gamma$, $\psi_2(q_D)=-2\pi/3-\gamma$ with weights $(c_{D,1},c_{D,2})=(-\eta,+\eta)$ and the oriented soliton angles $\phi_r^{(\epsilon)}$ $(r=1,2,3)$ with weights $(c_1,c_2,c_3)=(1,1,-2)$, define for any odd function $F(w)=-F(-w)$ of $w=\tan(\psi/2)$ the soliton and defect functionals **(c)**:

$$\mathcal{S}[F](\alpha,\epsilon):=\sum_{r=1}^{3}c_r\,\mathrm{Re}\,F\left(\tan\frac{\phi_r^{(\epsilon)}}{2}\right), \tag{34}$$

$$\mathcal{D}[F](q_D,\eta):=\sum_{s'=1}^{2}c_{D,s'}\,\mathrm{Re}\,F\left(\tan\frac{\psi_{s'}(q_D)}{2}\right)=\eta\left[\mathrm{Re}\,F\left(\tan\frac{\psi_2(q_D)}{2}\right)-\mathrm{Re}\,F\left(\tan\frac{\psi_1(q_D)}{2}\right)\right]. \tag{35}$$

Claim 4.1 (Universal KP-Coboundary Theorem, (c), verified in `kp_master_and_toda_edge.py`). *For every odd function $F(w)=-F(-w)$, the 3-root soliton functional (34) is the exact KP-chord difference of the 2-root defect dipole functional (35):*

$$\mathcal{S}[F](\alpha,\epsilon)=\mathcal{D}[F](p_b,+1)-\mathcal{D}[F](p_a,+1), \tag{36}$$

and along every edge $q_i\to q_{i+1}$ of any KP triple $\{q_1,q_2,q_3\}$ with soliton $S_i=(\alpha_i,\epsilon_i)$ on $\{q_i,q_{i+1}\}$ and sign $\rho_i=\eta\,\mathrm{sgn}(q_i-q_{i+1})$,

$$\mathcal{D}[F](q_{i+1},\eta)-\mathcal{D}[F](q_i,\eta)=\rho_i\,\mathcal{S}[F](S_i),\qquad\sum_{i=1}^{3}\rho_i\,\mathcal{S}[F](S_i)=0. \tag{37}$$

Proof. At $q_D=p_a=-\frac{1}{\sqrt{3}}\cos(\alpha+\epsilon\frac{\pi}{3})$ and $q_D=p_b=-\frac{1}{\sqrt{3}}\cos(\alpha-\epsilon\frac{\pi}{3})$, the two defect dipole angles $\psi_{1,2}(q_D)$ coincide (up to $\psi\mapsto-\psi$, which sends $w=\tan(\psi/2)\mapsto-w$ and flips the sign of any odd function $F(-w)=-F(w)$) with two of the three oriented soliton angles $\phi_r^{(\epsilon)}$. Specifically, the dipole weights $(-1,+1)$ at p_b assign $(+1,0,-1)$ to the three root differences of the KP triangle, while $(-1,+1)$ at p_a assign $(0,-1,+1)$; subtracting the two gives $(+1,0,-1)-(0,-1,+1)=(+1,+1,-2)=(c_1,c_2,c_3)$, establishing (36)–(37) for arbitrary odd $F(w)$. □

Notice that Claim 4.1 immediately unifies all one-loop single-soliton and defect-vacuum observables:

- Choosing $F_P(w) = -\frac{1}{2\pi}\frac{2w}{1+w^2}\ln|w|$ gives $\mathcal{S}[F_P] = \Delta P_{\mathrm{mode}}(\alpha,\epsilon)$ and $\mathcal{D}[F_P] = \Delta P_D(q_D,\eta)$;
- Choosing $F_E(w) = \frac{1}{2\pi}\frac{2w}{1+w^2}\big(1-\frac{1-w^2}{1+w^2}\ln|w|\big)$ gives $\mathcal{S}[F_E] = \Delta E_1^{\mathrm{ren}}(\alpha)$ and $\mathcal{D}[F_E] = \Delta E_D^{\mathrm{ren}}(q_D,\eta)$.

4.2 The two-vertex master function $\mathcal{F}_K(p,q)$ on the KP root space

Applying Claim 4.1 to each variable of an odd two-variable ratio kernel $K(w_1/w_2)$ ($K(-z) = -K(z)$) yields the fundamental building block of two-body soliton scattering on the KP root space.

Claim 4.2 (Two-Vertex Master Function $\mathcal{F}_K(p,q)$, (c), verified in `kp_master_and_toda_edge.py`). *For any odd kernel $K(z) = -K(-z)$, define the 2×2 dipole–dipole **two-vertex master function** on $(p,q)\in\mathbb{R}^2$ by*

$$\mathcal{F}_K(p,q) := \sum_{s_1=1}^{2}\sum_{s_2=1}^{2}(-1)^{s_1+s_2}\,\mathrm{Re}\,K\left(\tan\frac{\psi_{s_1}(p)}{2}\cot\frac{\psi_{s_2}(q)}{2}\right). \tag{38}$$

Then the 3×2 soliton–defect functional $\mathcal{I}_{SD}[K]$ is the first difference of $\mathcal{F}_K$,

$$\mathcal{I}_{SD}[K](\alpha,\epsilon;q_D,\eta) := \sum_{r=1}^{3}\sum_{s'=1}^{2}c_r c_{D,s'}\,\mathrm{Re}\,K\left(\tan\frac{\phi_r^{(\epsilon)}}{2}\cot\frac{\psi_{s'}(q_D)}{2}\right) = \eta\Big(\mathcal{F}_K(p_b,q_D)-\mathcal{F}_K(p_a,q_D)\Big), \tag{39}$$

and the 3×3 two-soliton functional $\mathcal{I}_{SS}[K]$ is the 4-vertex second difference (2-cocycle coboundary) of $\mathcal{F}_K$ across the soliton chords $(p_{1,a},p_{1,b})\times(p_{2,a},p_{2,b})$:

$$\begin{aligned}\mathcal{I}_{SS}[K](1,2) &:= \sum_{r=1}^{3}\sum_{s=1}^{3}c_r c_s\,\mathrm{Re}\,K\left(\tan\frac{\phi_{1,r}^{(\epsilon_1)}}{2}\cot\frac{\phi_{2,s}^{(\epsilon_2)}}{2}\right)\\ &= \mathcal{F}_K(p_{1,b},p_{2,b}) - \mathcal{F}_K(p_{1,a},p_{2,b}) - \mathcal{F}_K(p_{1,b},p_{2,a}) + \mathcal{F}_K(p_{1,a},p_{2,a}).\end{aligned} \tag{40}$$

*Consequently, for **every** odd kernel $K(z)$, $\mathcal{I}_{SD}[K]$ and $\mathcal{I}_{SS}[K]$ satisfy exact defect fusion and KP-triangle closure:*

$$\mathcal{I}_{SD}[K](3;q_{i+1},\eta) - \mathcal{I}_{SD}[K](3;q_i,\eta) = \rho_i\,\mathcal{I}_{SS}[K](3,S_i), \qquad \sum_{i=1}^{3}\rho_i\,\mathcal{I}_{SS}[K](3,S_i) = 0. \tag{41}$$

4.3 Finite-amplitude decomposition $I_{SS} = I_{SS}^{(\chi_2)} + I_{SS}^{(\ln)}$ and the small-amplitude corner

The one-loop kernel $K_L(z) = \frac{1}{2\pi}L(z)$ of Phase 2e and Phase 2g splits naturally into two odd kernels:

$$K_L(z) = K_{\chi_2}(z) + K_{\ln}(z), \qquad K_{\chi_2}(z) := -\frac{1}{\pi}\chi_2(z), \qquad K_{\ln}(z) := -\frac{1}{2\pi}\ln|z|\ln\left|\frac{1-z}{1+z}\right|\mathrm{sgn}(\mathrm{Re}\,z). \tag{42}$$

By Claim 4.2, both $I_{SS}^{(\chi_2)}(1,2) := \mathcal{I}_{SS}[K_{\chi_2}](1,2)$ and $I_{SS}^{(\ln)}(1,2) := \mathcal{I}_{SS}[K_{\ln}](1,2)$ are exact KP coboundaries at any finite amplitudes $(\alpha_1,\epsilon_1;\alpha_2,\epsilon_2)$, with $I_{SS}(1,2) = I_{SS}^{(\chi_2)}(1,2) + I_{SS}^{(\ln)}(1,2)$.

Claim 4.3 (Small-amplitude overtaking and head-on limits of $I_{SS}^{(\chi_2)}$ and $I_{SS}^{(\ln)}$, (c), verified in `kp_master_and_toda_edge.py`).

1. *In the small-amplitude overtaking corner* $\alpha_1 = r\lambda$, $\alpha_2 = \lambda \to 0^+$ *(*$\epsilon_1 = \epsilon_2 = +1$, $r \in (0,1)$*), the two KP coboundaries have the clean separation*

$$I_{SS}^{(\chi_2)}(r\lambda, +1; \lambda, +1) = -\frac{4}{\pi}\chi_2(r) + O(\lambda^2), \tag{43}$$

$$I_{SS}^{(\ln)}(r\lambda, +1; \lambda, +1) = -\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r} + O(\lambda^2) = -H_{\rm bad}(r) + O(\lambda^2). \tag{44}$$

Thus the logarithmic coboundary $I_{SS}^{(\ln)}$ *cancels the colliding Hirota quasi-momentum shift* $H_{\rm bad}(r)$ *at* $O(\lambda^0)$*, leaving the pure Legendre chi coboundary* $I_{SS}^{(\chi_2)}(r\lambda, \lambda) \to -\frac{4}{\pi}\chi_2(r)$ *(and* $-\frac{4}{\pi}(\chi_2(r) - r)$ *after the Fermi-edge subtraction at* $r \to 0$*).*

2. *In the small-amplitude head-on corner* $\alpha_1 = r\lambda$, $\epsilon_1 = +1$ *and* $\alpha_2 = \lambda$, $\epsilon_2 = -1$ *(*$\lambda \to 0^+$*),* $I_{SS}^{(\chi_2)}$ *and* $I_{SS}^{(\ln)}$ *individually carry opposite* $O(\lambda^2 \ln\lambda)$ *terms that cancel in their sum:*

$$I_{SS}^{(\chi_2)}(r\lambda, +1; \lambda, -1) = +\frac{3r}{\pi}\lambda^2\ln(r\lambda^2) + O(\lambda^2), \qquad I_{SS}^{(\ln)}(r\lambda, +1; \lambda, -1) = -\frac{6r}{\pi}\lambda^2\ln(r\lambda^2) + O(\lambda^2), \tag{45}$$

$$I_{SS}(r\lambda, +1; \lambda, -1) = \frac{r\lambda^2}{\pi}\left[-3\ln(r\lambda^2) + 6\ln 2 - \frac{3}{2}\ln 3 - 3\right] + O(\lambda^4\ln\lambda) \xrightarrow{\lambda\to 0^+} 0, \tag{46}$$

matching the vanishing $O(\hbar^0)$ *head-on residual of the quantum Toda Fermi-edge limit (*`matching_analysis.`

3. *At subleading order* $O(1/Y)$ *in the Toda Fermi-edge expansion* $R_0(Y, r) = -\frac{4}{\pi}(\chi_2(r) - r) + \frac{c_1(r)}{Y} + O(Y^{-2})$*, the two-loop KdV coefficient* $c_1(r)$ *extracted from* `matching_analysis.json` *is smooth and negative (*$c_1(0.4) = -0.007080$, $c_1(0.5) = -0.020991$, $c_1(0.6) = -0.048219$*).*

5 Summary of Claims, Failures Recorded, and Implications

Table 2 summarises all claims derived and verified in Phase 2h.

Failures recorded during the investigation

In accordance with `AGENTS.md`, we record all failed checks and sign/algebraic errors encountered and corrected during this task:

1. **Import path and ordering convention in `FastNodesEdgeBA`.** In the first run of `charge_and_frame_aud` we attempted to import `FastNodesEdgeBA` from `toda_edge_ba` rather than `matching_toda`, and wrote $r = \sqrt{u_1/u_2}$ ($u_1 < u_2$) instead of the `matching_toda.py` convention $y_1 = Y > y_2 = r^2 Y$ ($r = \sqrt{u_2/u_1} < 1$). Importing `FastNodesEdgeBA` from `matching_toda` and using $r = \sqrt{u_2/u_1}$ gave $\frac{\partial\Phi_{\rm cl}}{\partial u_1} = 2\ln\frac{1-r}{1+r} + 4r$ and $\frac{\partial\Phi_{\rm cl}}{\partial u_2} = -2\ln\frac{1-r}{1+r}$ with exact cancellation of $-\frac{2}{r}$ against $+\frac{2}{r}$.

2. **Missing $-\epsilon\sin\alpha\,\Delta P_{\rm mode}$ term when relating $\mathcal{A}_v(\alpha)$ to $-\frac{{\rm d}L_1}{{\rm d}\alpha}$.** When first coding `velocity_anomaly_cl` in `charge_and_frame_audit.py`, we equated $\mathcal{A}_v(\alpha) = \frac{{\rm d}\Delta E_1^{\rm ren}}{{\rm d}\alpha} - v\frac{{\rm d}\Delta P_{\rm mode}}{{\rm d}\alpha}$ directly to $-\frac{{\rm d}L_1}{{\rm d}\alpha}$, omitting the term $-\frac{{\rm d}v}{{\rm d}\alpha}\Delta P_{\rm mode} = +\epsilon\sin\alpha\,\Delta P_{\rm mode}$. In fact, $\mathcal{A}_v(\alpha)$ has light-root weight $\cos(2\alpha) - \cos^2\alpha = -\sin^2\alpha$, while $\epsilon\sin\alpha\,\Delta P_{\rm mode}$ has light-root weight $+\sin^2\alpha$; it is their sum $-\frac{{\rm d}L_1}{{\rm d}\alpha} = \mathcal{A}_v(\alpha) + \epsilon\sin\alpha\,\Delta P_{\rm mode}$ whose light-root weight $-\sin^2\alpha + \sin^2\alpha = 0$ vanishes identically.

3. **Sign convention of $(\Delta E_1^{\rm ren}, \Delta P_{\rm mode})$ and tadpole subtraction in quadrature check.** In an intermediate edit of `charge_and_frame_audit.py`, flipping the sign of $\Delta P_{\rm can}$ without flipping the Phase 2e/2g sign convention of $(\Delta E_1^{\rm ren}, \Delta P_{\rm mode})$, and using the un-cancelled polynomial counterterm $\omega(q) - q^2 - \frac{1}{2}$ instead of the exact Phase 2e/2g normal-ordering

Claim / Formula	Label	Verification Script	Precision / Result
Non-Lorentz-invariant velocity anomaly $\mathcal{A}_v(\alpha) = \frac{\mathrm{d}\Delta E_1^{\mathrm{ren}}}{\mathrm{d}\alpha} - v\frac{\mathrm{d}\Delta P_{\mathrm{mode}}}{\mathrm{d}\alpha} \neq 0$ (10) (Claim 2.1)	(c)	`charge_and_frame_audit.py`	Verified to 42+ digits across $\alpha \in (0, \pi/2)$ and $\epsilon = \pm 1$
Exact disappearance of light root $\phi_3 = \alpha$ in $\frac{\mathrm{d}L_1}{\mathrm{d}\alpha}$ (15), closed form of $\Delta P_{\mathrm{can}}(\alpha, \epsilon)$ (16) with zero $\alpha \ln \alpha$ term, and $\frac{\mathrm{d}\Delta E_{\mathrm{can}}}{\mathrm{d}v} = v\frac{\mathrm{d}\Delta P_{\mathrm{can}}}{\mathrm{d}v}$ (Claim 2.2)	(c)	`charge_and_frame_audit.py`	SymPy exact + numerical match $< 10^{-42}$; quadrature of $L_1(v)$ to $< 10^{-35}$
Overtaking amplitude-ordering reversal between good ($v = \cos\alpha$) and bad/Toda ($v = \cosh\beta$) equations and sign flip $H_{\mathrm{bad}}(r) = -H_{\mathrm{good}}(r)$ (21)	(c)	`charge_and_frame_audit.py`	SymPy exact series identity
Quantum Fermi-edge quasi-momentum shift $u_j = y_j - \frac{1}{2\pi}\ln y_j + c_{\mathrm{edge}}$ (26) and exact cancellation of $-2/r$ Darboux singularity by $+4\sqrt{u_1 u_2}$ in $\frac{\partial \Phi_{\mathrm{cl}}}{\partial u_2}$ (29) (Claim 3.1)	(c)	`charge_and_frame_audit.py`	SymPy exact derivatives + `FastNodesEdgeBA` Richardson check
Cancellation of $\pm 4r\delta u_1$ and $\pm\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ in $R_0(Y, r) = d\Phi_{\mathrm{cl}} + d\Phi_{\mathrm{loop}}$ and exact limit $\lim_{Y\to\infty} R_0(Y, r) = -\frac{4}{\pi}(\chi_2(r) - r)$ (33) (Claim 3.2)	(c)	`charge_and_frame_audit.py`	Matches `matching_tables.tex` to 1.2–7.0×10^{-5} (slope to 2.9×10^{-4})
Universal KP-Coboundary Theorem $\mathcal{S}[F](\alpha, \epsilon) = \mathcal{D}[F](p_b, +1) - \mathcal{D}[F](p_a, +1)$ for arbitrary odd $F(w)$ (36)–(37) (Claim 4.1)	(c)	`kp_master_and_toda_edge.py`	Tested on 5 odd functionals across 12 KP triangles: max residual $< 10^{-43}$
Two-vertex master function $\mathcal{F}_K(p, q)$ (38), first difference $= \mathcal{I}_{SD}[K]$ (39), second difference $= \mathcal{I}_{SS}[K]$ (40), and universal KP-triangle closure (Claim 4.2)	(c)	`kp_master_and_toda_edge.py`	Tested on 4 odd kernels ($K_L, K_{\chi_2}, K_{\ln}, K_{\mathrm{rat}}$): max residual $< 10^{-43}$
Finite-amplitude split $I_{SS} = I_{SS}^{(\chi_2)} + I_{SS}^{(\ln)}$, small-amplitude overtaking cancellation $I_{SS}^{(\ln)} + H_{\mathrm{bad}} = O(\lambda^2)$ (43)–(44), and head-on limit (45)–(46) (Claim 4.3)	(c)	`kp_master_and_toda_edge.py`	Richardson small-λ residual $< 10^{-16}$

Table 2: Summary of claims and verification checks in Phase 2h.

tadpole subtraction in the quadrature check of $L_1(v)$, produced $O(1)$ mismatches (0.380 and 0.149). Restoring the exact Phase 2e/2g definitions reduced all residuals to $< 10^{-42}$ (and 0.0 at 50-digit quadrature).

4. **Head-on small-amplitude coefficients of $I_{SS}^{(\chi_2)}$ and $I_{SS}^{(\ln)}$.** In the first run of Check 3c of `kp_master_and_toda_edge.py`, we wrote $-\frac{6}{\pi}r\lambda^2$ for the small-amplitude head-on limit of $I_{SS}(r\lambda, +1; \lambda, -1)$, overlooking the leading logarithmic term $-\frac{3}{\pi}r\lambda^2\ln(r\lambda^2)$ coming from $I_{SS}^{(\chi_2)} = +\frac{3}{\pi}r\lambda^2\ln(r\lambda^2) + O(\lambda^2)$ and $I_{SS}^{(\ln)} = -\frac{6}{\pi}r\lambda^2\ln(r\lambda^2) + O(\lambda^2)$ (recorded in Claim 4.2 of `oneloop_phase.pdf`). Testing against the exact asymptotic law (46) gave agreement to $< 10^{-8}$.

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